

TURKISTAN, Kazakhstan

WMO: 381980

Lat: 43.2719N

Lon: 68.1914E

Elev: 679

StdP: 14.34

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	4.7	-7.6	3.7	1.6	-1.3	5.2	7.1	29.6	28.4	21.5	27.1	2.2	50	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	103.2	66.9	100.3	66.0	97.8	65.4	69.3	95.3	68.2	94.0	67.1	92.4	7.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	85.2	74.2	59.5	77.8	74.9	57.4	72.1	75.6	33.8	94.7	32.8	93.7	31.9	92.5	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	22.3	17.2	13.9	DB	-4.3	108.1	9.8	2.7	-11.4	110.0	-17.1	111.6	-22.6	113.1	-29.7	115.1
(5)				WB	-4.8	72.0	9.7	2.2	-11.8	73.5	-17.5	74.8	-22.9	76.0	-30.0	77.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	27.6	32.2	47.4	59.8	71.5	80.2	83.2	81.4	70.0	55.1	41.9	29.6
	DBStd	21.76	10.82	11.79	9.62	8.76	6.73	6.20	5.91	6.53	7.77	8.94	9.72	10.41
	HDD50	2340	695	501	164	21	0	0	0	0	1	52	274	632
	HDD65	4997	1160	918	548	199	20	1	0	1	37	323	694	1096
	CDD50	4816	0	4	83	314	665	906	1029	973	602	209	31	0
	CDD65	2000	0	0	2	42	221	457	564	509	189	15	1	0
	CDH74	27025	0	1	31	465	2550	6318	8248	6971	2206	229	6	0
	CDH80	14900	0	0	3	124	1081	3588	5000	4074	990	40	0	0
Wind	WSAvg	5.2	4.1	4.8	5.6	6.7	6.7	5.8	5.5	6.0	5.2	4.2	3.6	3.6
Precipitation	PrecAvg	8.0	0.8	0.9	1.1	1.1	0.9	0.3	0.1	0.0	0.1	0.4	0.9	1.1
	PrecMax	15.8	1.9	2.9	3.1	2.8	2.9	1.8	0.9	0.7	0.7	2.2	2.2	2.7
	PrecMin	4.1	0.2	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.3	0.4	0.6	0.6	0.8	0.8	0.4	0.2	0.1	0.1	0.5	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	62.9	77.7	88.4	96.9	104.5	108.2	106.2	98.7	84.8	74.1	54.8
		MCWB	45.2	50.2	56.5	62.2	64.4	67.6	67.7	67.5	64.2	58.6	55.7	45.6
	2%	DB	48.7	56.3	72.1	83.5	92.8	100.9	104.2	102.1	93.9	79.7	66.0	49.7
		MCWB	42.2	46.5	54.7	60.2	62.9	65.9	67.1	66.7	62.7	56.4	52.8	43.6
	5%	DB	44.1	51.5	66.9	79.6	89.6	98.0	100.7	99.4	90.0	76.1	60.3	46.0
		MCWB	39.3	44.2	52.6	58.7	61.7	65.4	66.3	65.9	61.1	55.2	49.3	41.2
	10%	DB	40.5	47.0	62.2	75.3	86.2	95.2	97.8	96.6	85.8	71.7	56.1	42.6
		MCWB	37.1	40.9	50.8	56.9	61.0	64.6	65.7	64.8	59.2	53.2	47.2	39.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.6	52.0	59.7	65.6	67.7	70.2	71.5	70.7	66.2	60.7	57.9	47.9
		MCDB	52.5	60.6	70.2	80.3	90.1	95.2	95.8	98.6	94.2	80.0	68.7	51.8
	2%	WB	43.0	48.1	56.7	62.9	65.7	68.4	69.5	68.8	63.5	58.2	54.6	44.5
		MCDB	47.4	55.1	68.6	77.4	85.6	93.0	95.6	96.3	90.2	75.1	64.0	48.2
	5%	WB	40.2	44.6	54.3	60.8	64.2	67.3	68.3	67.4	61.9	56.3	51.0	41.8
		MCDB	43.6	49.8	64.4	75.3	83.2	91.7	94.2	94.9	87.0	72.7	59.2	45.7
	10%	WB	37.4	41.6	52.0	58.7	62.8	66.1	67.3	65.9	60.2	54.2	47.8	39.0
		MCDB	40.3	46.4	60.9	72.1	80.6	90.0	92.8	93.3	84.1	69.6	54.4	42.6
Mean Daily Temperature Range	5% DB	MDBR	13.8	16.3	20.0	20.9	23.4	25.2	25.5	25.8	25.9	24.3	17.8	14.0
		MCDBR	17.0	21.9	25.8	26.2	26.8	27.6	27.4	28.1	29.1	28.9	23.6	18.1
		MCWBR	12.1	13.5	12.9	11.1	9.6	9.5	9.0	9.6	11.4	13.6	13.4	12.7
	5% WB	MCDBR	15.0	19.1	22.0	23.2	23.8	25.6	25.4	27.0	27.5	26.1	20.6	15.7
		MCWBR	11.4	12.9	12.3	10.8	9.4	9.1	9.0	9.9	11.6	13.0	12.3	11.6
Clear-Sky Solar Irradiance	taub	0.360	0.393	0.451	0.481	0.466	0.451	0.436	0.403	0.380	0.407	0.402	0.357	
	taud	2.058	2.033	1.967	1.938	2.022	2.093	2.145	2.222	2.239	2.078	2.052	2.085	
	Ebn at Noon	229	244	246	250	258	262	264	270	267	236	211	217	
	Edh at Noon	34	42	51	57	54	50	47	42	38	40	35	31	
All-Sky Solar Radiation	RadAvg	519	840	1220	1722	2187	2429	2389	2156	1694	1088	631	446	
	RadStd	71	83	94	161	131	101	95	68	67	75	65	51	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+1.60	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	+1.75	N/A	N/A	N/A	N/A	+329	+239

Nomenclature: See separate page